AN EXPLICIT 5-REGULAR TWO-STAGE COUNTEREXAMPLE TO THE GRAPH-THEORETIC BENEŠ CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. The graph-theoretic form of the Beneš conjecture asserts that if L is a simple d-regular ordered two-stage graph such that L^m is externally connected for some $m \geq 1$, then L^{2m} is rearrangeable. Following Chojecki (2026) at $N = 5$ and $m = 2$, we verify an explicit 5-regular ordered two-stage graph L_5 on 10 vertices per layer with 50 directed edges such that L_5^2 is externally connected (reaching all 100 source-target pairs), while L_5^4 is not rearrangeable: the 5-regular cross-exchange permutation mask requires 25 edge-disjoint paths from $X = \{x_0, \ldots, x_4\}$ to $Y = \{y_0, \ldots, y_4\}$, whereas the four-stage network has minimum s-t cut capacity $16 < 25$.

1. Main Counterexample

Let $V = X \sqcup Y$ with $X = \{x_0, \ldots, x_4\}$ and $Y = \{y_0, \ldots, y_4\}$, and define the 50 directed edges of L_5 by $x_0 \to y_j$, $x_i \to x_j$ $(1 \leq i \leq 4)$, $y_0 \to x_j$, and $y_i \to y_j$ $(1 \leq i \leq 4)$ for all $0 \leq j \leq 4$.

Theorem 1.1. *The simple* 5*-regular ordered two-stage graph L_5 satisfies:*

(1) *In two stages, every $u \in V$ reaches every $v \in V$ ($x_i \to x_0 \to y_j$ and $x_i \to x_1 \to x_j$, and symmetrically from Y), so L_5^2 is externally connected ($F(L_5) = 2$).*

(2) *In the four-stage network L_5^4, only the* 5 *edges $x_0 \to y_j$ per stage cross from X to Y (*20 *edges in total across* 4 *stages), and the layered capacity network from X at stage* 0 *to Y at stage* 4 *has minimum s-t cut capacity* 16*. Consequently, the* 5*-regular request mask demanding* 5 *parallel paths $x_i \to y_i$ and $y_i \to x_i$ (*25 *paths from X to Y) is unroutable in L_5^4.*

Date: August 27, 2026.

2020 *Mathematics Subject Classification.* Primary 05C20; Secondary 68R10, 90B10.